

AN-SDR BASED MULTI-CHANNEL FM TRANSMITTER

Daniel Phutinyane^{a,*}, Monageng Kgwadi^a, Ecco D. Maje^b

^a*University of Botswana, Gaborone, Botswana*

^b*Botswana Institute for Technology Research and Innovation, Gaborone, Botswana*

Abstract

Frequency modulation (FM) transmission still remains an effective way of broadcasting radio content to audiences globally. FM signals are known to have short propagation distances and as such, achieving wider footprint requires the use of radio repeaters/transmitters. This poses a challenge for broadcasters in the developing world due to the prohibitive cost associated with installing and maintaining those transmitters. Software-defined radio (SDR) offers a low-cost solution by performing in software most of the signal processing traditionally carried out on dedicated hardware. In addition, the behaviour of an SDR-based transmitter can be further customised and/or configured to the needs of each particular broadcaster, a flexibility which allows for the broadcast of multiple channels from one common SDR hardware platform. This paper demonstrates a low-cost multiple channel FM transmitter with up to three channels on a Nuand BladeRF 2.0 micro xA4 software defined radio. All the three channels were successfully received using commercial FM radio receivers and this success leads to the elimination of

*Corresponding author

Email address: phutinyaned@gmail.com (Daniel Phutinyane)

multiple transmitters in one transmitter site, which would otherwise consume more power. It also allows cost sharing on installation and maintenance fees and as a result, broadcasters can afford to further set up more transmitter sites hence a wider footprint. Moreover, the setup can be used in disaster recovery to restore services rapidly after natural disasters.

Keywords: Software Defined Radio, Frequency Modulation, GNU Radio, BladeRF

1. Introduction

The concept of frequency modulation (FM) was realized by Edwin Howard Armstrong, who invented “wideband” FM radio in 1933 [1]. This technology, involves varying the frequency of the carrier signal to convey information from a radio station to the listener. Since its invention, there has been significant improvement of radio technology mostly because of improvements in the electronic components used [2]. To date, FM radio is still the most widely used means of receiving radio broadcast worldwide [3].

Generally, broadcasters worldwide implement an FM broadcast system by deploying a satellite downlink, an integrated receiver decoder (IRD), the radio data system (RDS), an exciter, amplifiers, combiners (diplexer) and an antenna. An IRD connected to a low noise blocker (LNB) and satellite dish receives and converts the radio signal from radio frequency (RF) signal to base-band. The output goes through RDS for the unique station identifiers to be decoded. It is then sent to an exciter, where the baseband signal is mixed with a carrier frequency generated by a crystal oscillator. The result

is a low power frequency modulated signal that is fed to an amplifier. The amplifier then amplifies the signal from the exciter before it can be sent to the antenna via a diplexer unit (combiner). The multiplexing of these channels in a diplexer is done such that the harmonics from the transmitted channels do not interfere with each other. Common practice is that they should fade off within few meters from the transmitter site. The space between two adjacent channels is normally 200KHz [4], which is made up of 75KHz each side of the carrier plus 25KHz guard band on either side. Figure 1 shows a two channel FM transmitter.

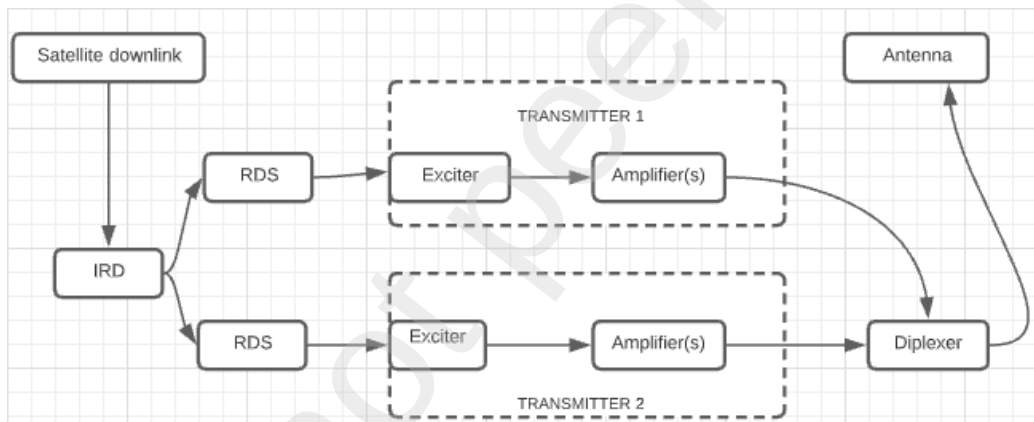


Figure 1: A Typical FM Transmission Block Diagram showing a setup that deploys two transmitters

Each radio station requires one set of this broadcast transmitter system for their respective FM channel. Due to the propagation characteristics of FM signals (limited by distance) [5] [6], comprehensive coverage of large areas requires multiple installations. This has so far limited the penetration of FM in remote areas (in developing countries). In Botswana, each of the three private radio stations, namely; Yarona FM, Gabz FM and Duma FM, have

only managed to install twenty three (23) transmitters across the country [7]. This paper presents a cost effective alternative using SDR to broadcast multiple channels in one SDR hardware, which to the best of our knowledge, has never been done before. By sharing resources, broadcasters can reduce their capital and maintenance costs, and in cases of disaster, services can be restored quickly.

2. Background

Frequency modulation transmission consists of mono audio (L+R) at 15 kHz as well as the stereo audio (L-R) [8] obtained by amplitude modulation of the baseband using double sideband suppressed carrier onto a 38-KHz carrier [9]. A pilot tone at 19KHz is also transmitted for coherent demodulation of the DSB-SC signal at the receiver side. Finally, the "RDS signal is inserted above the stereo multiplex using a subcarrier frequency of 57 kHz" [10] to allow for transmission of services like station identification and automatic tuning of channels etc. Figure 2 illustrates the typical FM stereophonic spectrum of a broadcast.

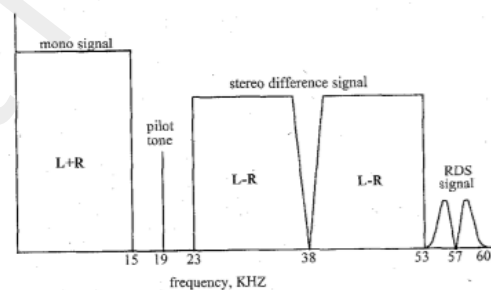


Figure 2: Spectrum of a stereophonic programme with RDS. [9]

FM broadcast evolved from the use of balanced reactance tubes [11], ca-

capacitance variation of tubes (e.g., the Miller effect), or mechanical devices (e.g., the BW-5 TV sideband analyzer) to semiconductor devices, particularly capacitive diodes [12] and variable divider in phase locked loops [13], through dual output digital exciter (which consists of two local oscillators [14]). All these inventions had their shortcomings, such as sensitivity to vibrations, limited controlled range, circuit complexity, and the necessity to reduce the carrier modulation so as not to exceed a phase departure of $\pm 90^\circ$ [12].

Software Defined Radio (SDR) have emerged to address the challenges brought by its predecessors, and its flexibility has presented new opportunities which can be exploited to address limitations of old techniques. The term, Software Defined Radio was first coined by Joe Mitola in 1991 to refer to reprogrammable or reconfigurable radios [15]. In this system, some radio functionalities are performed by software. Advances in electronics have made it possible to define by software typical functionality of a radio interface usually implemented by dedicated hardware [16]. The signal processing (modulation/demodulation) is carried out in software instead of a dedicated hardwired circuitry which allows greater flexibility of building circuitry to handle different types of radio signals. Furthermore, the same SDR hardware can be programmed to optimally adapt to its environment for effective use of the spectrum using cognition cycle [17] to create a cognitive radio [18].

Several commercially available SDR platforms were considered in this study as shown in Table 1, extracted from a list at [20].

Table 1: Features of some of the common SDR hardwares [20]–[23]

Properties	Crimson TNG	ADALM PLUTO	BladeRF 2.0 micro xA4	AFEDRI SDR-Net
Tuning Range	110 MHz- 6.8 GHz	70MHz- 6000MHz with soft- ware modi- fication	47MHz- 6GHz	35MHz- 1700 MHz
Max RF bandwidths	upto 322 MHz for each TX chain	20MHz	56 MHz per channel	2.3MHz
Transmit power(dBm)	18	7	+15.6	no trans- mitter capabilities
Cost(US\$)	13,500.00	148	480	249

The overall cost paired alongside some other features gave Blade RF Micro 2.0 xA4 an edge over the rest. For example, whilst Adalm Pluto was the cheapest of all the mentioned SDRs, it was disadvantaged by its maximum RF bandwidth of 20MHz compared to the 56MHz of BladeRF, as well as its transmit power of 7 dBm. Using the same argument, it was found that AFEDRI SDR-Net can only work as a receiver. Meanwhile Crimson TNG's tuning range is out of range of the FM band region II (87.5-108 MHz) [24]

as well as its relatively high cost of \$13,500.00.

Regarding SDR softwares, there are various options available on the market that can be used alongside the SDR hardware, and this study considered Matlab/Simulink, Labview, and GNU Radio. Matlab/Simulink and LabView requires license subscription whilst GNU Radio is free and as result, it became a preferred choice.

3. Experimental Set Up

The Blade RF Micro 2.0 SDR platform was programmed with the software to broadcast three channels described previously. The output of the SDR was amplified via a Nuand BT-100 bias tee power amplifier. Reception was done by tuning the FM receiver on a cellphone to the three channels (104.0 MHz, 105.0 MHz, 106.0 MHz) and the signal was clearly received. Figure 3 shows the experimental setup with the cellphone tuned to the station broadcasting on 105.0 MHz.

4. Experimental Results and Analysis

These results were obtained by varying various parameters of the GRC software shown in Figure 4. Each of these blocks served a distinct role towards realization of multi-channel FM broadcasting as itemized below.

- **WAV FILE SOURCE:** This block takes in a path for a stored audio file and allows it to be loaded as an input into the flowgraph. Three

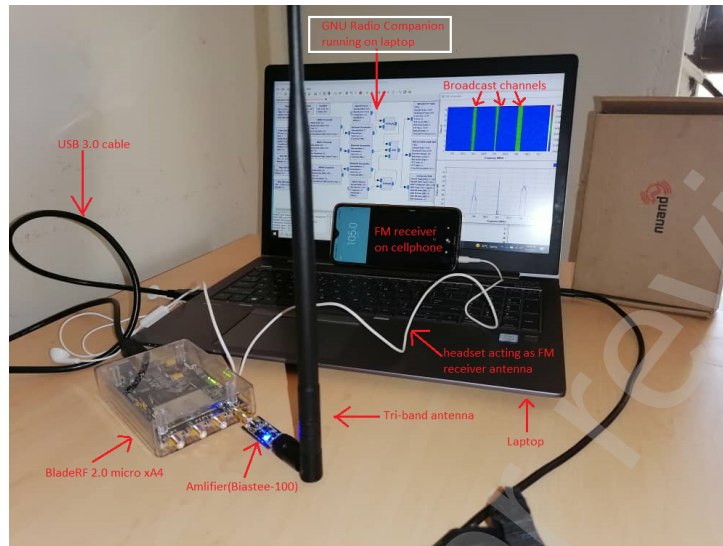


Figure 3: FM reception test using cellphone radio and headsets as antenna. All the three channels were received by tuning to their respective frequencies.

different wave (.wav) formatted files were each loaded into three wav file source blocks to emulate three radio stations. Each of the wav files took the place of the radio feed from the satellite link. In practical deployment, audio streams from three separate radio channels can be used instead of local files.

- WBFM Transmit: A wide-band frequency modulating (WBFM) block whose function is to modulate an audio into a frequency modulated signal. Three of these WBFM transmit blocks were employed to modulate each of the three respective base-band audio feeds into a FM modulated signals. In the WBFM block, the audio rate was set to 36 KHz and the quadrature rate was set at a multiple frequency of audio rate of 360 KHz. The other parameters were left at their default values,

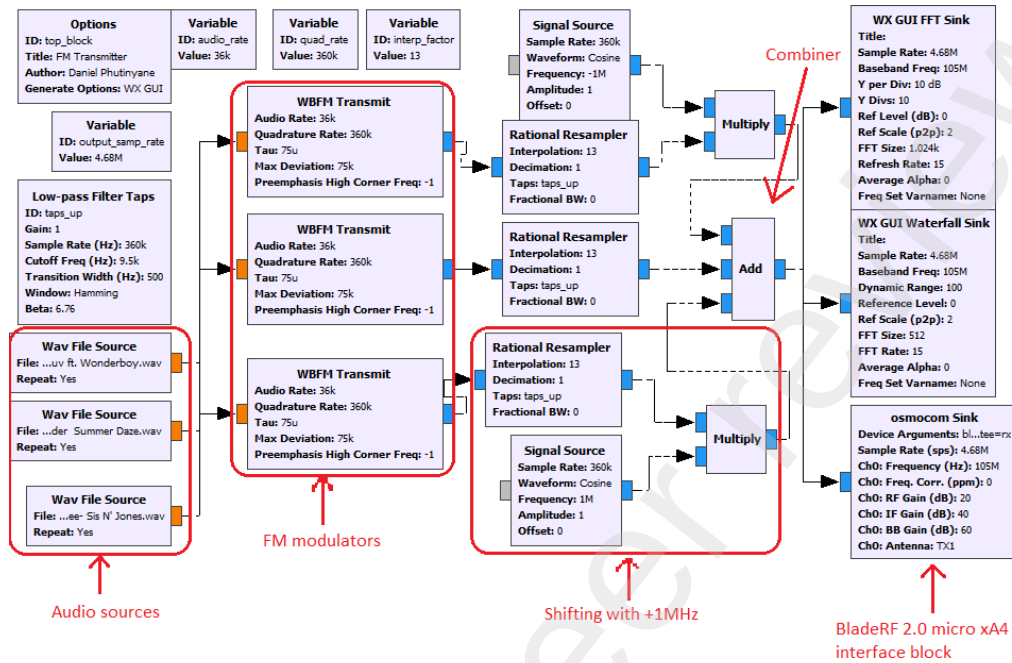


Figure 4: GRC flowchart for three FM transmitters running on BladeRF 2.0 micro xA4.

with maximum deviation parameter being consistent with the 75 KHz for wide-band FM.

- **RATIONAL RE-SAMPLER:** A rational re-sampler block was used to up-sample the base-band from 360 KHz to a preferred sample rate of 4.5 MHz which was determined by listening to the quality of the received audio at the receiver side. This was achieved by interpolating by a factor of 13. To filter out the high frequency components, a low pass with a cut-off frequency obeying Nyquist rate of 9.5 KHz was used for each channel. This was achieved by calling a variable taps_up which is a function stored in LOW PASS FILTER TAPS block.
- **SIGNAL SOURCE:** Depending on the selected waveform, this block

generated a cosine waveform with arbitrarily chosen frequencies (± 1 MHz). This frequency determines how much from the centre frequency will the other channels shift.

- **MULTIPLY:** This block was used in connection with the signal source to shift two of the three outputs from the three WBFM blocks to a pre-determined frequency. The multiplication operation on the two of the streams shifted the channels to distinct bands within the FM range. A multiplication with $+1$ MHz signal produced a shift to the left whilst -1 MHz shifted to the right. The ability to select an arbitrary frequency to shift the modulated FM signal allows flexibility and agility throughout the FM broadcasting range, 87.0 MHz to 108.0 MHz.
- **ADD:** The samples were then combined together by add block to produce an output consisting of all the three channels multiplexed into one stream.
- Finally, the graphical representation of the output of the SDR was plotted using Fast Fourier Transform (FFT) sink blocks. It visualizes the output of the modulated signal whose outputs are depicted in figure 5 showing the three distinct channels at 104.0 MHz, 105.0 MHz and 106.0 MHz.

The unfiltered channels as expected, are wider than the filtered channels owing to the higher frequency components which introduces noise to the transmitted signal. The difference between the rising edge (105.813 MHz) and falling edge (106.174 MHz) at a consistent power level of -60.0858 dB of a randomly selected channel with centre frequency of 106.0 MHz, determined

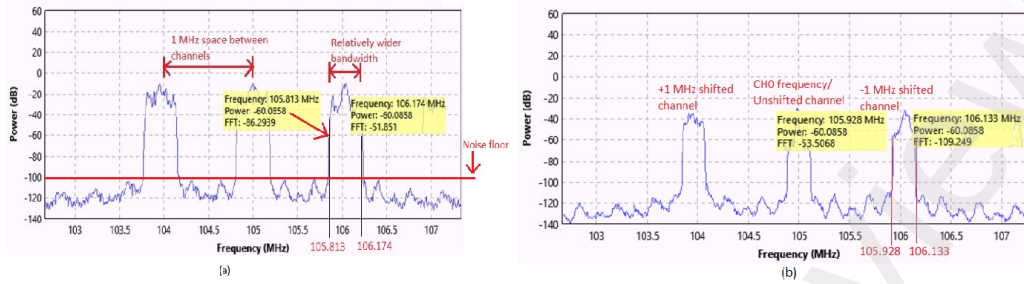


Figure 5: (a) Unfiltered and (b) filtered diagrams for the multiplexed wideband FM channels showing a 361 KHz and 200 KHz bandwidth respectively.

the bandwidth to be 0.361 MHz as shown in figure 5a. This value was higher than the expected FM bandwidth of 0.2 MHz (200 KHz) which is the limit for FM radio broadcast [25]. The received channel on the FM receiver was also noisy. The selection of power level was arbitrary but it was maintained consistently on either side of the spectrum to get the true value of the frequency at either side.

Meanwhile, figure 5b shows the results of a filtered channel where each channel was limited to a bandwidth of 200 KHz. This is observed by following the rising and falling edge at 105.928 MHz and 106.133 MHz respectively which yielded a bandwidth of 0.205 MHz. The low pass filter called by taps_up variable (as shown in figure 7b) limited each channel to occupy only 9.5 KHz and avoiding interference with adjacent channels. This limiting of the channel bandwidth is important for co-existence to avoid interference.

Bypass of filtering, to achieve figure 5a, was achieved by leaving the taps field empty under the Rational Resampler block. This is shown in Figure 6 below.

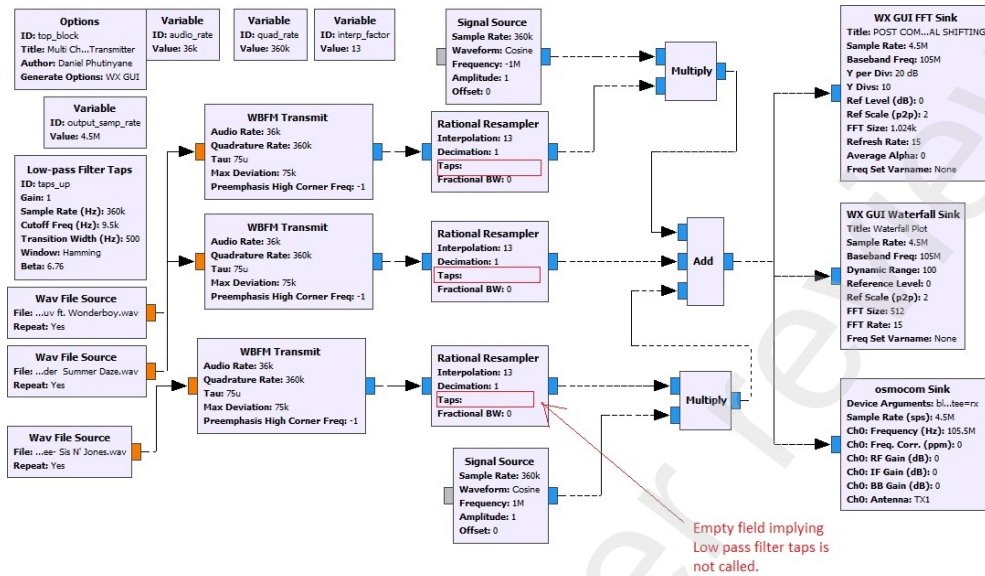


Figure 6: GRC flowchart for three (3) FM channels. The taps field has been left empty so as to bypass low pass filter.

Flexibility is an important feature for co-existence and the SDR allows re-configuration and/or system customization. For example, in Botswana there is currently an existing radio station broadcasting at 103.0 MHz in Gaborone. Applying a frequency shift of +2 MHz to a centre frequency of 105.0 MHz would imply the shifted channel is moved to 103.0 MHz which would interfere with the existing channel. To overcome this, the channel centred at 105.0 MHz was changed to 105.5 MHz as shown in Figure 8.

Furthermore, by setting the frequency of the signal source block as -1 MHz and +2 MHz (Figure 8) the other two (2) channels were shifted to the right (106.5 MHz) and left (103.5 MHz) respectively as shown in Figure 9a. These results were confirmed by tuning to those frequencies on the FM radio

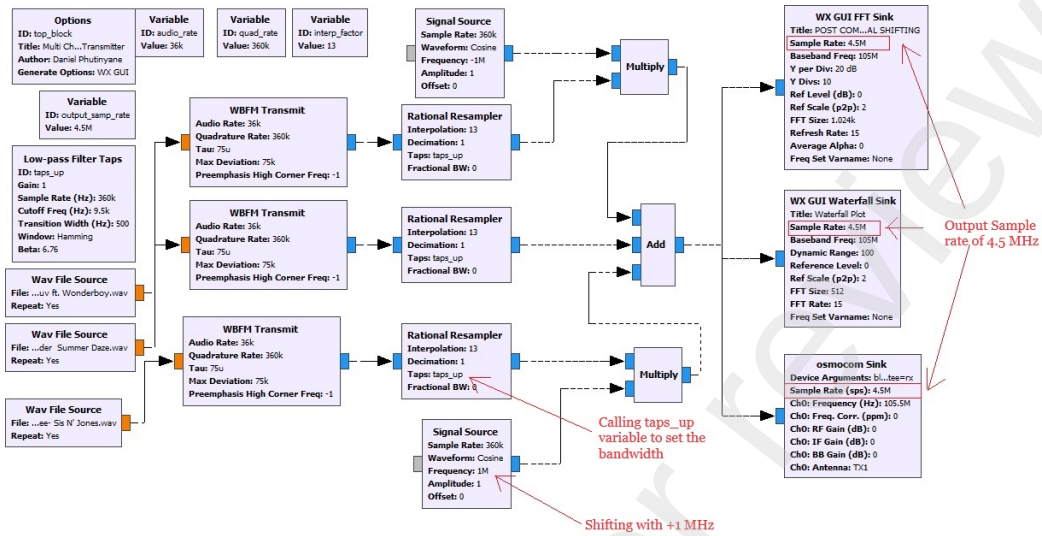


Figure 7: GRC flowgraph for three FM transmitters running on BladeRF 2.0 micro at a sample rate of 4.5 MHz with low-pass filter taps.

receiver, and since the space between the already existing channel of 103.0 MHz was greater than the minimum required of 200 KHz, there was no interference heard on either 103.0 MHz or 103.5 MHz.

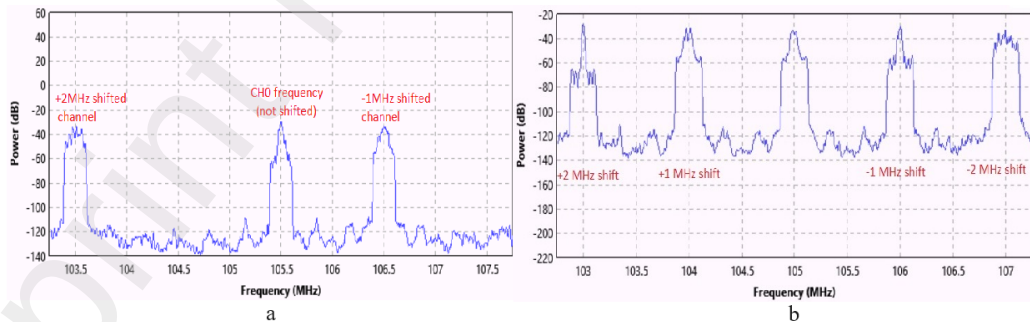


Figure 9: a) FFT of three (3) channels obtained by performing +2MHz and -1MHz frequency shift of a local oscillator. b) FFT showing five (5) channels.

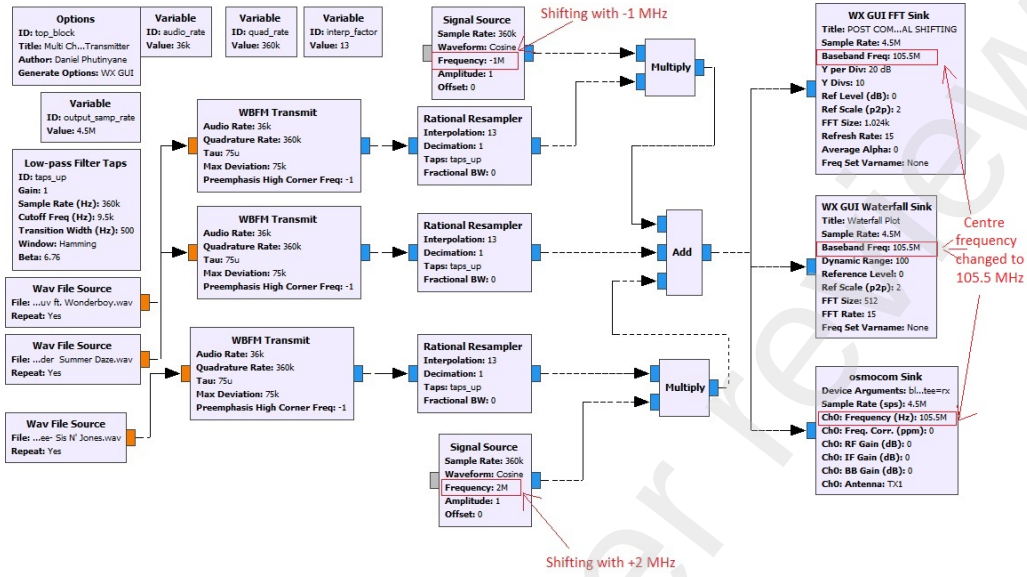


Figure 8: GRC flowchart of three arbitrary broadcast channels obtained by performing +2 MHz and -1 MHz frequency shift of a channel centred at 105.5 MHz

Through experiments, it was also found that up to five channels can be obtained although with a lower audio quality than for the three channels. Figure 9b shows the spectrum of these channels.

4.1. Verification and Testing

Experimental validation of the multi-channel FM transmitter was done in the unlicensed low-power FM transmission which is limited to 100 mW (20 dBm) effective radiated power. With the antenna gain rated at 5 dBi and the bias-tee at 15 dB, the RF gain from the device was set to 0 dB (through the RF gain parameter under the Osmocom sink block). This implies that Equivalent Isotropically Radiated Power; $EIRP = AntennaGain(dB) +$

$TransmitterPower(dB) = 5dBi + (15dB + 0dB) = 20dBi$. This power is not split between the three (3) channels. Rather, the antenna "sees" the combined signal as one and radiates it in that form.

5. Conclusions

Simultaneous transmission of three FM signals from one SDR was successfully realised. By applying an arbitrary frequency shift of ± 1 MHz to a signal at a centre frequency of 105 MHz, there were two other channels at 104.0 MHz and 106.0 MHz. Similarly, two other channels were received at 103.5 MHz and 106.5 MHz when performing -1 MHz and +2 MHz shift to a CH0 frequency of 105.5 MHz. These frequencies were successfully received on the already available FM receivers without interference. For practical purposes, inclusion of RDS is required as well as an amplifier with higher gain to cover a larger area. Furthermore, additional work can be done to establish the relationship between sample rate and shifting factor with regard to position of the output spectrum.

6. Acknowledgements

This research would have not been possible without the financial grant by the Office of Research and Development at the University of Botswana. My sincere thanks also goes to Mr Bathopi Luke-Director Broadcasting Regulation at Botswana Communications Regulatory Authority (BOCRA), whose input as an industry expert played a significant role towards realisation of this project.

References

- [1] G. L. Frost, *Early FM radio: Incremental technology in twentieth-century America*. JHU Press, 2010.
- [2] S. Maas, “Armstrong and the superheterodyne: A historical look at the mixer,” *IEEE Microwave Magazine*, vol. 14, no. 6, pp. 34–39, 2013.
- [3] T. Hussain, T. Mujahid, M. Abdullah, S. Saeed, and S. A. Alharbi, “Sd-drm: Software defined digital radio mondiale,” in *2019 International Conference on Electrical, Communication, and Computer Engineering (ICECCE)*, IEEE, 2019, pp. 1–5.
- [4] P. Bogere, R. Akol, and J. Butime, “Optimization of frequency modulation band for terrestrial radio broadcasting: The case of uganda,” in *2015 IEEE International Conference on Microwaves, Communications, Antennas and Electronic Systems (COMCAS)*, 2015, pp. 1–5. DOI: 10.1109/COMCAS.2015.7360389.
- [5] A. Popteev, “Improving ambient fm indoor localization using multipath-induced amplitude modulation effect: A year-long experiment,” *Pervasive and Mobile Computing*, vol. 58, p. 101022, 2019.
- [6] M. G. Domingo, P. K. E. Estebal, G. A. Tongco, and G. P. Mappatao, “Development of a reliable path-loss model for fm broadcast reception in office locations,” *Bulletin of Electrical Engineering and Informatics*, vol. 9, no. 4, pp. 1654–1661, 2020.
- [7] *Bocra20*, <https://www.bocra.org.bw/sites/default/files/documents/BOCRA20.pdf>, (Accessed on 06/21/2021).

- [8] *Mono vs stereo sound: What's the big difference?* - rowkin, <https://www.rowkin.com/blogs/rowkin/mono-vs-stereo-sound-whats-the-big-difference>, (Accessed on 07/10/2021).
- [9] C. S. Koukourlis, J. E. Plevridis, and J. N. Sahalos, "A new digital implementation of the rds in the fm stereo," *IEEE transactions on broadcasting*, vol. 42, no. 4, pp. 323–327, 1996.
- [10] C. Roppel and A. O. Perez, "An efficient implementation of an fm/rds software radio," in *4th European Education and Research Conference (EDERC 2010)*, IEEE, 2010, pp. 245–249.
- [11] M. G. Crosby, *Reactance tube modulation*, US Patent 2,349,811, May 1944.
- [12] A. Bott, "New concept in fm exciters," *IEEE Transactions on Broadcasting*, no. 1, pp. 22–25, 1965.
- [13] W. B. Bruene, *Fm broadcast exciter apparatus*, US Patent 3,902,019, Aug. 1975.
- [14] G. Cabrera, T. W. Dittmer, and W. Douglas-Duella, *Dual output digital exciter*, US Patent 7,623,833, Nov. 2009.
- [15] J. H. Reed, *Software radio: a modern approach to radio engineering*. Prentice Hall Professional, 2002.
- [16] E. Buracchini, "The software radio concept," *IEEE Communications Magazine*, vol. 38, no. 9, pp. 138–143, 2000.

- [17] W. Song, "Configure cognitive radio using gnu radio and usrp," in *2009 3rd IEEE International Symposium on Microwave, Antenna, Propagation and EMC Technologies for Wireless Communications*, 2009, pp. 1123–1126. DOI: 10.1109/MAPE.2009.5355934.
- [18] J. Mitola and G. Maguire, "Cognitive radio: Making software radios more personal," *IEEE Personal Communications*, vol. 6, no. 4, pp. 13–18, 1999. DOI: 10.1109/98.788210.
- [19] I. Martoyo, P. Setiasabda, H. Y. Kanalebe, H. P. Uranus, and M. Pardede, "Software defined radio for education: Spectrum analyzer, fm receiver/transmitter and gsm sniffer with hackrf one," in *2018 2nd Borneo International Conference on Applied Mathematics and Engineering (BICAME)*, IEEE, 2018, pp. 188–192.
- [20] *List of software-defined radios - wikipedia*, https://en.wikipedia.org/wiki/List_of_software-defined_radios, (Accessed on 05/30/2024).
- [21] *Crimson tng - per vices — software defined radio*, <https://www.everythingrf.com/products/software-defined-radios/per-vices/957-1642-crimson-tng>, (Accessed on 05/30/2024).
- [22] *Adalm-pluto sdr active learning module*, <https://www.analog.com/media/en/news-marketing-collateral/product-highlight/ADALM-PLUTO-Product-Highlight.pdf>, (Accessed on 05/30/2024).
- [23] *Afedri sdr-net description*, <https://web.archive.org/web/20130103172850/http://www.afedri-sdr.com/>, (Accessed on 05/30/2024).

- [24] *Fm / tv regional frequency assignment plans*, <https://www.itu.int/en/ITU-R/terrestrial/broadcast/Pages/FMTV.aspx>, (Accessed on 05/30/2024).
- [25] L. Der, "Frequency modulation (fm) tutorial," *Silicon Laboratories Inc*, 2008.